Chongqing Family Translator — Technical Architecture & Design

Oct 7, 2026 · @Khoon Seng

Overview

A voice-first web app turns a Mandarin-speaking tour guide into live Cantonese audio for five family members, and their replies into Mandarin for the guide, over Singapore roaming data in Chongqing.

Only two of the seven adults speak Mandarin. The app removes the need for them to interpret every conversation by hand.

Goals

- Speech is the main input; typing is available as a fallback.
- Every family listener hears Cantonese audio; the guide hears Mandarin audio.
- Large-font text sits beside the audio as a check on meaning.
- Works anywhere in Chongqing on SG roaming or a non-local-breakout eSIM.
- Field-tested and ready in about two months.

Non-goals

- Offline translation, or working on domestic Chinese networks and hotel Wi-Fi.
- Native iOS or Android apps; this is a mobile web app (PWA).
- Simultaneous interpretation. Consecutive interpretation, a few seconds behind the speaker, is the target.

Success criteria

- Translated audio starts within roughly 2–4 seconds of the speaker finishing a sentence.
- The in-laws pick the Cantonese voice and translations in a blind test.
- No bubble is lost after a dead zone or reconnect.
- Daily running cost stays in single-digit US dollars.

Users, languages and channels

The app needs only two audio channels, Cantonese and Mandarin, plus text in three languages.

Person	Speaks	Hears (audio)	Reads (text)	Device
Guide	Mandarin	Mandarin, from the hub speaker	Simplified Chinese on the hub	Shared hub phone
In-laws	Cantonese	Cantonese (must-have, audio first)	Colloquial Cantonese, Traditional characters, large font	Own phone + earphones
Wife's siblings	Cantonese, English	Cantonese (must-have)	Cantonese or English	Own phone + earphones
Khoon Seng and wife	Mandarin, English, Cantonese	None needed	Optional	Operate the hub; human fallback

Input language is chosen explicitly on the hub with **Guide** and **Family** talk buttons, because auto-detecting Mandarin versus Cantonese on short clips is unreliable. Family input can be Cantonese or English.

The Cantonese text is written as colloquial spoken Cantonese (粵語口語, e.g. 佢哋喺度等緊我哋), not formal written Chinese, so the same string reads naturally on screen and sounds natural when spoken.

Hub phone and audio capture

The hub is Khoon Seng's own phone with no extra hardware; a clip-on mic is deferred until field tests show it is needed.

Modes

- **Guide** is a latch: one tap starts continuous listening, and voice-activity detection splits the guide's speech at natural pauses.
- To let family speak, tap **Family**; tap **Guide** again to return. The whole screen is tinted a different colour per mode so the current one is unmistakable.

- In Guide mode, segments the STT flags as not Mandarin, or as low confidence, are dropped. This only filters out family chatter; it never chooses the pipeline.

Placement

- In the car: phone in a dashboard or vent mount, bottom edge (main mic) toward the guide, radio off.
- On foot: phone held at chest height, within about 1 m of the guide.

Hub phone rules

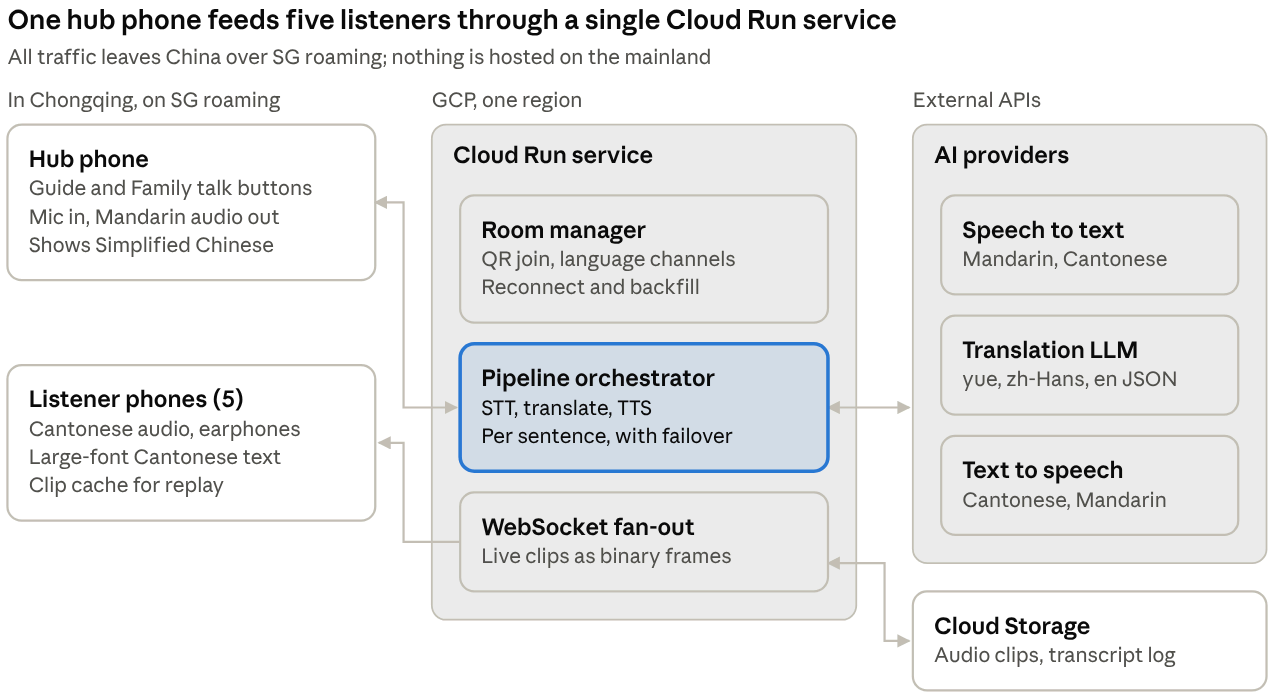
- No Bluetooth earbuds connected to the hub phone; the phone may silently switch to their mic.
- Bring a car charger and power bank: continuous mic, screen wake lock and roaming data drain the battery fast.

Clip-on mic (deferred)

Decided at the weeks 6–7 field test by comparing transcripts: phone held at 1 m versus a clip-on worn by the person playing the guide. A clip-on only helps on the guide, not on the hub operator. If bought, it is a wireless kit whose receiver plugs into USB-C or Lightning, never Bluetooth, and the app simply uses the phone's default audio input.

System architecture

A single Cloud Run service sits between one shared hub phone and five listener phones, calling external STT, translation and TTS APIs on their behalf.



system architecture · hub, Cloud Run, providers, listeners

The hub sends speech in and plays Mandarin out; the orchestrator runs each sentence through the providers, stores the clip in Cloud Storage and pushes it to listeners over WebSockets.

Mermaid version, for the Markdown copy in the repo:

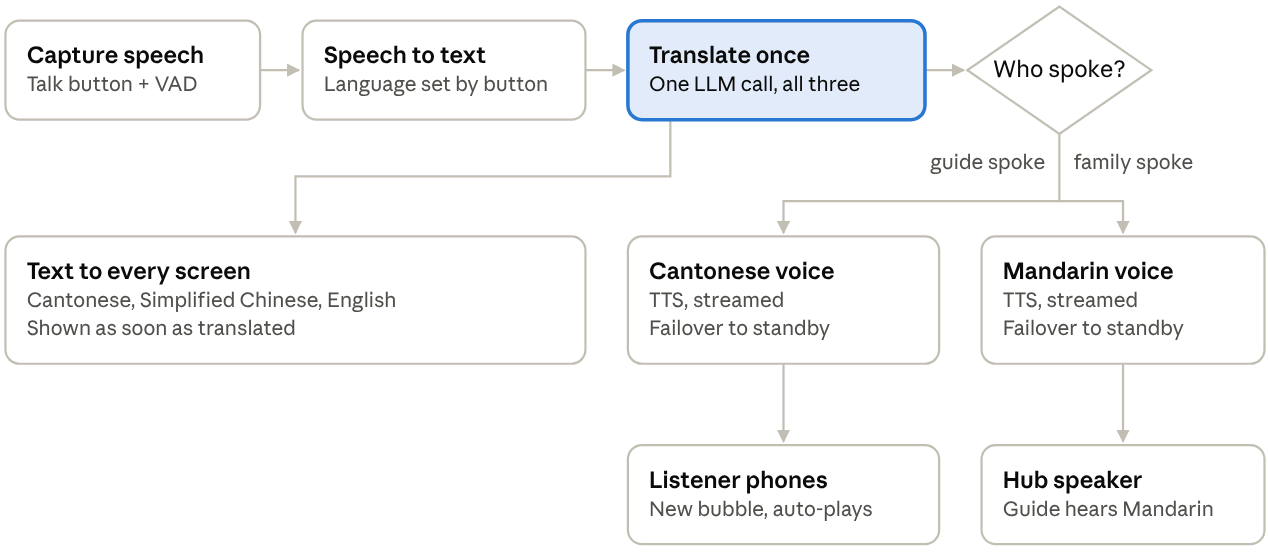
```
flowchart LR
  subgraph CQ["In Chongqing, on SG roaming"]
    HUB["Hub phone<br/>Guide / Family modes<br/>Mic in, Mandarin audio
out<br/>Simplified Chinese on screen"]
    LIS["Listener phones (5)<br/>Cantonese audio, earphones<br/>Large-font
Cantonese text<br/>Clip cache for replay"]
  end
  subgraph GCP["GCP, one region: Cloud Run service"]
    ROOM["Room manager<br/>QR join, language channels<br/>Reconnect and
backfill"]
    ORCH["Pipeline orchestrator<br/>STT, translate, TTS<br/>Per sentence,
with failover"]
    FAN["WebSocket fan-out<br/>Live clips as binary frames"]
  end
  subgraph API["External APIs: AI providers"]
    STT["Speech to text<br/>Mandarin, Cantonese"]
    LLM["Translation LLM<br/>yue, zh-Hans, en JSON"]
```

```
    TTS["Text to speech<br/>Cantonese, Mandarin"]
  end
  GCS[("Cloud Storage<br/>Audio clips, transcript log")]
  HUB <--> ORCH
  ORCH <--> STT
  ORCH <--> LLM
  ORCH <--> TTS
  ORCH --> FAN
  ROOM --- FAN
  FAN --> LIS
  FAN <--> GCS
```

Translation pipeline

Every sentence runs through one cascade: speech to text, a single translation call, then text-to-speech only for the listeners who need audio.

Each sentence is translated once, then voiced only for those who need it



translation pipeline · per sentence

Guide speech becomes Cantonese audio for the listeners; family speech becomes Mandarin audio from the hub speaker. Every screen gets text in all three languages.

Mermaid version, for the Markdown copy in the repo:

```
flowchart TD
  CAP["Capture speech<br/>Guide latch or Family, plus VAD"] --> STT["Speech
to text<br/>Language set by mode"]
  STT --> TR["Translate once<br/>One LLM call: yue, zh_hans, en"]
  TR --> TXT["Text to every screen<br/>Cantonese, Simplified Chinese,
English"]
  TR --> WHO{"Who spoke?"}
  WHO -- guide spoke --> YUE["Cantonese voice<br/>TTS, streamed<br/>Failover
to standby"]
  WHO -- family spoke --> CMN["Mandarin voice<br/>TTS, streamed<br/>Failover
to standby"]
  YUE --> LP["Listener phones<br/>New bubble, auto-plays"]
  CMN --> HS["Hub speaker<br/>Guide hears Mandarin"]
```

A single cascade replaces a hybrid of OpenAI's realtime translator plus a Cantonese cascade. One pipeline keeps the Cantonese and Mandarin channels in step and leaves one integration to maintain.

Model and provider choices

Every stage sits behind a small adapter interface, and the winners are picked by a week-1 bake-off judged by the in-laws.

Stage	Candidates	Standby / fallback	Chosen by
STT, Mandarin (guide)	OpenAI gpt-4o-mini-transcribe, Whisper large-v3 on Groq (paid tier), Google Cloud STT	Any of the others	Accuracy on Chongqing-accented Mandarin, noisy street audio
STT, Cantonese (family)	Google Cloud STT (Cantonese), Azure Speech zh-HK, OpenAI gpt-4o-transcribe	Any of the others	Accuracy on the in-laws' own recordings, incl. Cantonese–English code-switching
Translation LLM	A fast frontier-tier model: GPT mini class, Gemini Flash, Qwen	Second model from the list	Natural colloquial Cantonese output; no Mandarin grammar in Cantonese characters

Stage	Candidates	Standby / fallback	Chosen by
TTS, Cantonese	Google Chirp 3 HD yue-HK (Preview, streaming, available in asia-southeast1)	Azure zh-HK neural voices (GA); Qwen3-TTS Cantonese voices	In-laws' blind listening test
TTS, Mandarin	Same providers as Cantonese	Same	Clarity for the guide

Translation call design. One LLM call per sentence returns all three targets as JSON (`yue`, `zh_hans`, `en`), streamed so TTS can start on the first sentence. The prompt carries a glossary of Chongqing place and dish names (解放碑, 洪崖洞, 磁器口, 毛血旺) and only the last 2–3 utterances as context.

Failover. If a provider errors or is too slow on a sentence, the adapter retries that sentence on the standby. As a last resort the app shows the text and flags the hub operator to interpret.

Ruled out. OpenAI GPT-Realtime-Translate (no Cantonese output voice), conversational realtime models (an assistant, not an interpreter), Groq TTS (no Chinese voices) and Groq's free tier (request caps run out within hours of continuous use).

Listener experience and playback rules

The listener screen works like a WhatsApp voice-note thread: audio first, large-font Cantonese text under every bubble, and any bubble replayable on its own.

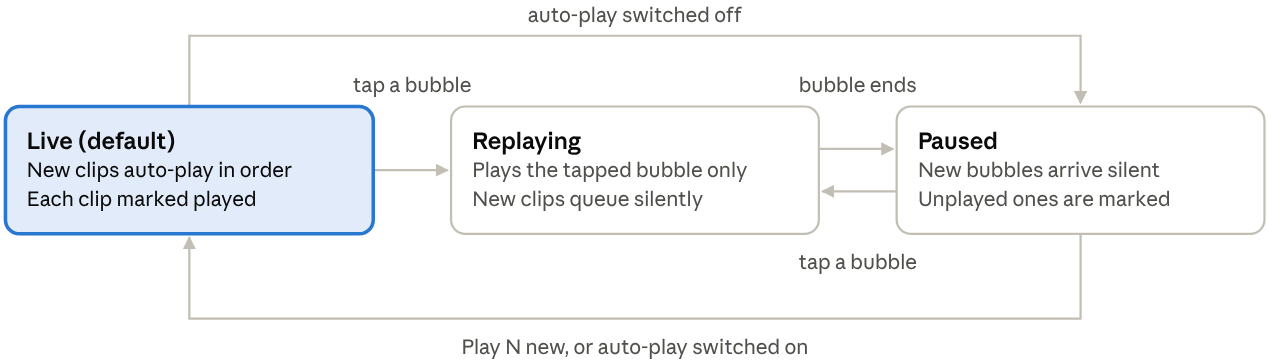
Bubbles

- One bubble per speaker turn; each sentence inside it is its own audio clip.
- Tap a bubble to replay the whole turn; tap a sentence to replay just that sentence.
- The guide's original Mandarin shows in small text under each bubble, so Khoon Seng or his wife can check a confusing translation on the spot.
- Unplayed bubbles carry a visible marker, like unheard voice notes.

Player states

A replay never continues on its own; new clips wait until the listener chooses

Listener player states; Live is the default when a phone joins



listener player states, Live, Replaying, Paused

A replay plays only the tapped bubble and then stops in Paused; it never runs on into later bubbles. In Paused, new clips arrive silently so the listener can read instead of listen, until they tap **Play N new** or switch auto-play back on.

Mermaid version, for the Markdown copy in the repo:

```
stateDiagram-v2
  [*] --> Live
  Live --> Replaying: tap a bubble
  Replaying --> Paused: bubble ends
  Paused --> Replaying: tap a bubble
  Replaying --> Replaying: tap another bubble (current one stops)
  Live --> Paused: auto-play switched off
  Paused --> Live: Play N new, or auto-play switched on
  note right of Live: New clips auto-play in order, each marked played
  note right of Replaying: Plays the tapped bubble only, new clips queue
silently
  note right of Paused: New bubbles arrive silent, unplayed ones are marked
```

- Tapping another bubble mid-replay stops the current one at once; two clips never overlap.
- Replays never re-call TTS: clips are cached on the phone (IndexedDB) and backed by Cloud Storage, so a replay works even in a tunnel.
- Each phone shows a tap-to-start screen on joining, which unlocks audio on iOS, and holds a screen Wake Lock while in a room.

Hosting, networking and China access

The roaming SIM, not the cloud region, is what keeps the app reachable in Chongqing; the region is picked by a latency test.

Connectivity

- Every phone uses SG roaming or an eSIM confirmed to route outside China (no local breakout). Traffic exits in Singapore, so Google, OpenAI and Groq behave as they do at home.
- Wi-Fi stays off while the app is in use. Hotel and café Wi-Fi is on the domestic network, where Google domains are blocked whatever the region.
- Nothing is hosted in mainland China, which would need an ICP filing.

Compute: Cloud Run

- One service holds the Vue PWA, the FastAPI WebSocket endpoint and the pipeline, built as one container.
- `max-instances=1`, because room state lives in memory and a second instance would split the room.
- WebSocket connections are cut at the request timeout (60 minutes maximum); the client's reconnect-and-backfill logic handles it.
- Run at 1 vCPU. Below 1 vCPU, Cloud Run forces maximum concurrency to 1, which would allow only one WebSocket connection.
- Provider API keys live server-side in Secret Manager; phones never call providers directly.

Region

- Candidates: `us-central1` (inside the Cloud Storage free tier) and `asia-southeast1` (closest to the roaming exit and to Google's Cantonese voices).
- In week 1, deploy to both and measure time-to-first-audio from a Singapore phone. Keep the faster one.
- Cloud Run's free tier is a monthly spending discount per billing account, so it applies in either region.

Storage: Cloud Storage

- Each sentence's audio is saved as an Opus clip, plus a transcript log per room.
- Live clips travel inline over the WebSocket as binary frames; the bucket serves backfill after reconnects and catch-up fetches.

- A full trip is a few hundred MB, inside the 5 GB free allowance in US regions and pennies elsewhere.

Joining a room. The hub creates a room and shows a QR code holding a short-lived room token. Listeners scan it once, pick their language and stay joined for the day.

Latency, rate limits and cost

The target is about 2–4 seconds from end of sentence to translated audio, at a few US dollars a day; tokens never pile up because every call is stateless.

Latency budget (approximate, to be measured in week 1)

Step	Expected time
STT on one sentence	0.5–1.5 s
Translation LLM, first sentence streamed	0.5–1 s
TTS, first audio bytes	0.5–1 s
Roaming path and region hop	0.3–0.6 s

Rate limits

- Each STT, translation and TTS request stands alone, so there is no growing context to exhaust. Translation passes only the last 2–3 utterances.
- Groq's free tier caps Whisper at 20 requests per minute and 2,000 per day, and its LLMs at about 1,000 requests per day. Continuous touring exhausts these within hours, so any Groq use needs the paid Developer plan.
- Three design choices keep request counts low: voice-activity detection on the phone so silence is never sent, sentence-sized chunks instead of fixed 2–3 second windows, and one LLM call returning all three languages.

Cost (approximate, for about 3 hours of speech a day)

Item	Per day
STT (per-minute pricing, e.g. about US$0.003/min for gpt-4o-mini-transcribe)	Under US$1
Translation LLM	Under US$0.50

Item	Per day
TTS for two voices	A few US dollars
Cloud Run and Cloud Storage	Near zero; WebSockets may exceed the free vCPU allowance by a dollar or two for the trip

Prices were checked against public pricing pages in October 2026 and should be re-checked before the bake-off.

Risks and mitigations

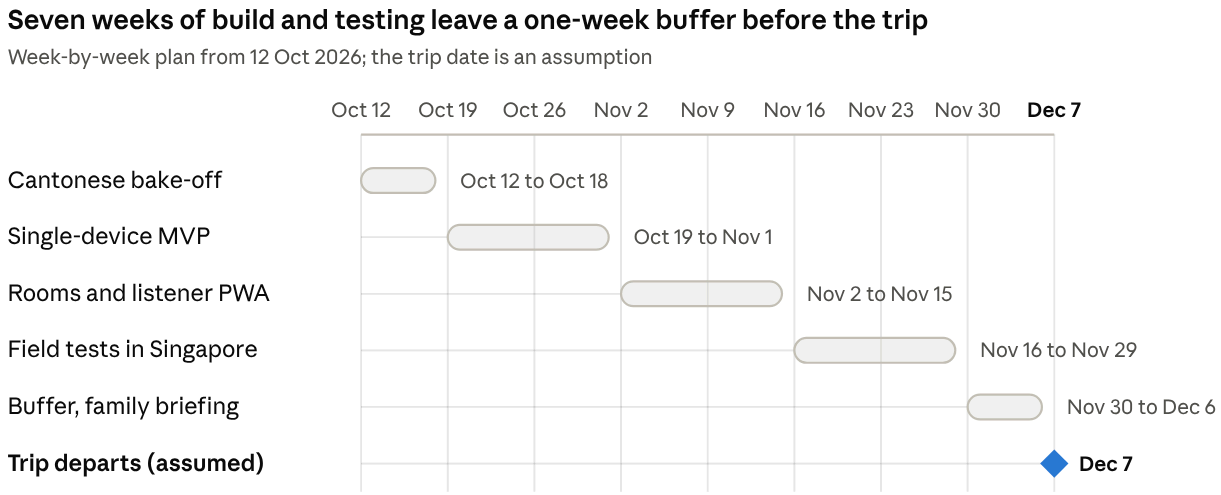
The biggest risks are audio capture and Cantonese quality, not hosting.

Risk	Impact	Mitigation
eSIM uses local breakout in China	App unreachable	Use SG roaming or a verified eSIM; keep a Qwen/Alibaba provider path as plan B
Phone joins hotel Wi-Fi	App blocked	Wi-Fi off at start of day; app shows a clear connection banner
Street noise, guide far from the hub	Poor STT	Hold the phone within about 1 m of the guide; add a clip-on mic for the guide only if the field test shows it is needed
Family chatter while Guide mode is latched	Junk bubbles transcribed as Mandarin	Listeners on earphones; drop segments flagged as not Mandarin or low confidence
Hub phone battery drains	Translation stops mid-tour	Car charger and power bank
Bluetooth earbuds connected to the hub	Phone silently switches to the earbud mic	Keep the hub phone free of Bluetooth audio devices
Chongqing-accented Mandarin	STT errors	Ask the guide for standard Putonghua; test on Chongqing-accented samples

Risk	Impact	Mitigation
Translation slips into Mandarin grammar	Cantonese sounds wrong to the in-laws	Colloquial-Cantonese prompt; blind bake-off of models
iOS audio unlock and screen lock	Listener hears nothing	Tap-to-start on join; Wake Lock API; test on the in-laws' actual phones
Tunnels and dead zones	Missed bubbles	Reconnect and backfill from the last utterance id
Cantonese TTS still in Preview at Google	Voice outage	Adapter failover to Azure zh-HK; last resort text plus human interpreting
Mandarin vs Cantonese auto-detection	Wrong pipeline	Explicit Guide and Family modes on the hub
Listener phones playing aloud	Echo and feedback into the hub mic	Earphones for every listener

Eight-week build plan

Seven weeks of build and field testing start on 12 October, leaving the final week as buffer before an assumed early-December departure.



eight-week build plan, by week

Mermaid version, for the Markdown copy in the repo:

```
gantt
  title Eight-week build plan (trip date assumed)
  dateFormat YYYY-MM-DD
  axisFormat %d %b
  section Build
  Cantonese bake-off         :a1, 2026-10-12, 7d
  Single-device MVP          :a2, 2026-10-19, 14d
  Rooms and listener PWA     :a3, 2026-11-02, 14d
  section Test
  Field tests in Singapore   :a4, 2026-11-16, 14d
  Buffer and family briefing :a5, 2026-11-30, 7d
  section Trip
  Trip departs (assumed)     :milestone, m1, 2026-12-07, 0d
```

1. **Week 1, Cantonese bake-off.** Run 10–15 recorded Mandarin narration sentences through 2–3 translation models and 2–3 Cantonese voices, blind, for the in-laws to judge. Score each Cantonese STT on their recorded questions. Deploy a test service to both regions and confirm the eSIM's routing.
2. **Weeks 2–3, single-device MVP.** The hub records, shows text and plays audio, with provider adapters in place.
3. **Weeks 4–5, rooms and listener PWA.** QR join, WebSocket fan-out, bubbles, player states, on-phone cache, reconnect and backfill.
4. **Weeks 6–7, field tests in Singapore.** The family uses their own phones in noisy places such as hawker centres and the MRT, with someone acting as the Mandarin guide. Compare transcripts from the phone held at 1 m against a clip-on worn by the "guide" to decide whether to buy a mic.
5. **Week 8, buffer.** Fixes, then briefing the family on joining a room and using replay.

Open decisions

Seven decisions remain; most close after the week-1 bake-off. Decided on 7 Oct: the hub is the owner's phone, with Guide mode latching into continuous listening.

- [ ] Trip dates and the final roaming plan or eSIM provider
- [ ] Whether the guide will be driving or seated separately (affects phone placement in the car)
- [ ] Cloud Run region: `us-central1` or `asia-southeast1`, after the latency test
- [ ] Winning STT, translation model and Cantonese voice, after the bake-off

- [ ] New-bubble cue while paused: vibration on Android, chime or animation on iOS
- [ ] Clip-on mic for the guide: only if the weeks 6–7 field test shows phone-only is not enough
- [ ] Whether English text shows by default for the wife's siblings